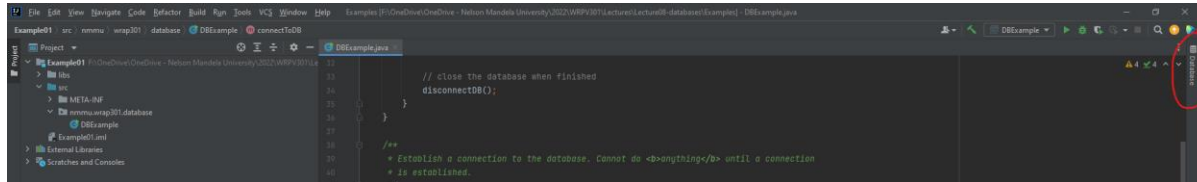
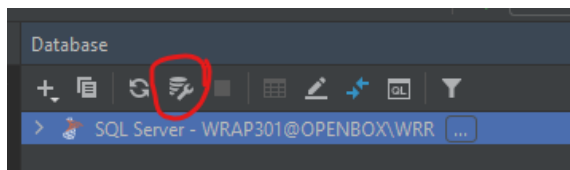


The first thing you need to do is to ensure you are connected to the university's VPN (if you are connecting from home)

We'll be changing Lecture08's Example01 in this scenario. So once you've opened in IntelliJ, you can click the Database button in the top right:



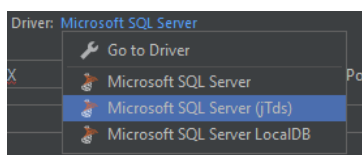
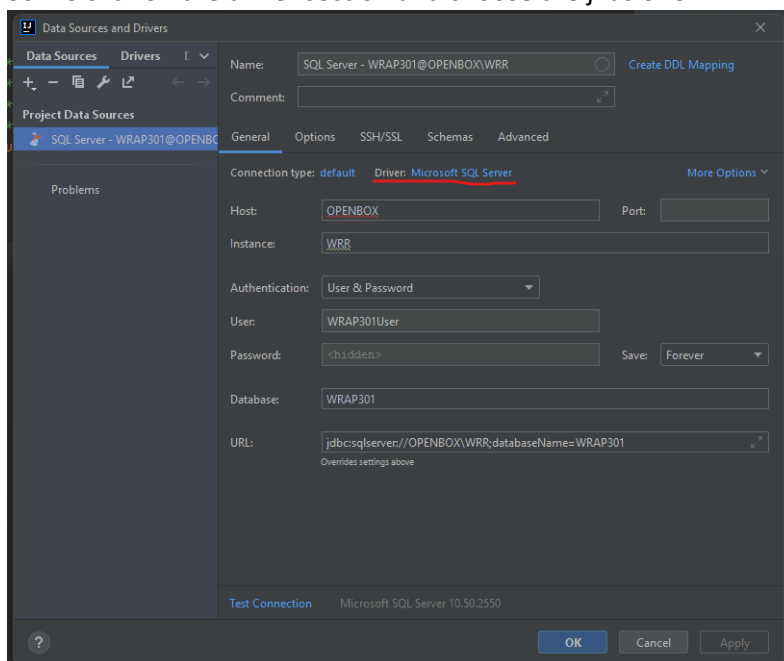
It should have the DB listed there from when Dieter was making the example. Press the “Data Source Properties” button:



When connecting from home, you need to give the full Host name. So instead of just OPENBOX, you must use openbox.nmmu.ac.za. However, the new DB is hosted on postgresql.mandela.ac.za as far as I know. So use that preferably 😊.

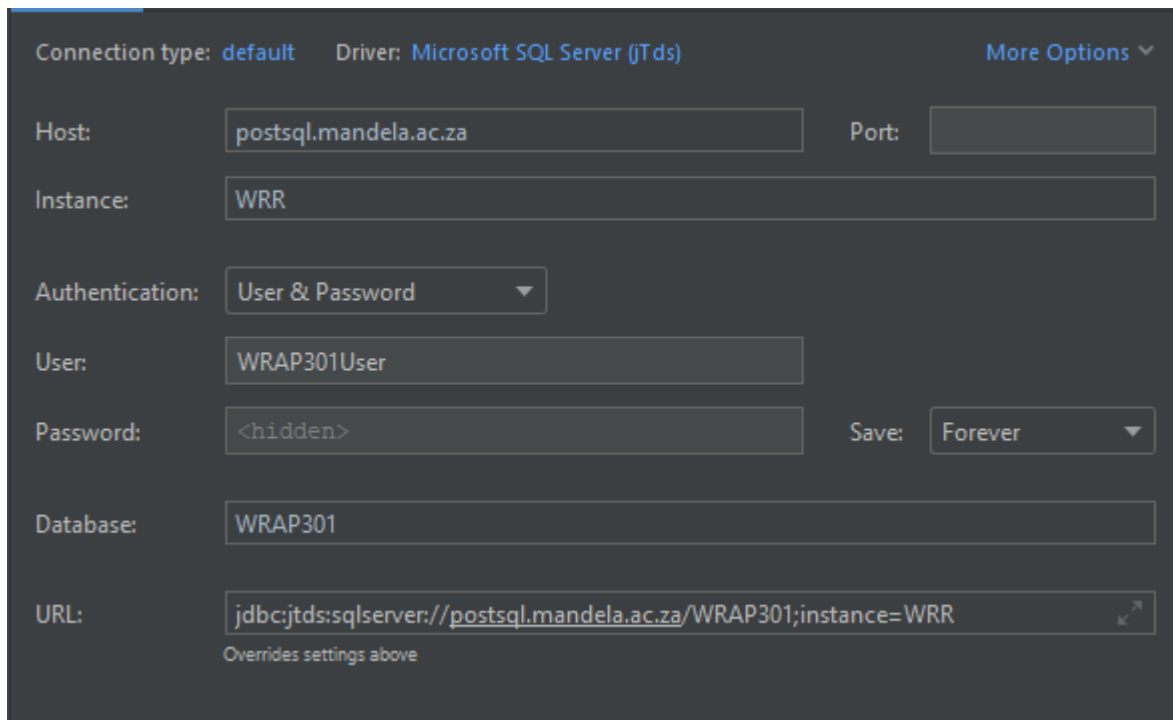
The default Microsoft SQL Server driver is causing some issues (due to security patches or something like that), so to circumvent that, we will change the driver to rather use the Microsoft SQL Server (jTds) driver (I don't know why it works with this driver, but it does 🤪).

So we click on the driver section and choose the jTds one.



This might require you to download this particular driver, so just accept the download if IntelliJ asks.

So your Data Sources and Drivers window should look something like this:



The screenshot shows the 'Data Sources' configuration window in IntelliJ IDEA. At the top, it says 'Connection type: default' and 'Driver: Microsoft SQL Server (jTds)'. There is a 'More Options' dropdown. The fields are as follows:

- Host: postgresql.mandela.ac.za
- Port: (empty)
- Instance: WRR
- Authentication: User & Password (dropdown)
- User: WRAP301User
- Password: <hidden>
- Save: Forever (dropdown)
- Database: WRAP301
- URL: jdbc:jtds:sqlserver://postgresql.mandela.ac.za/WRAP301;instance=WRR

Below the URL field, it says 'Overrides settings above'.

Note that the driver is jTds and the fields:

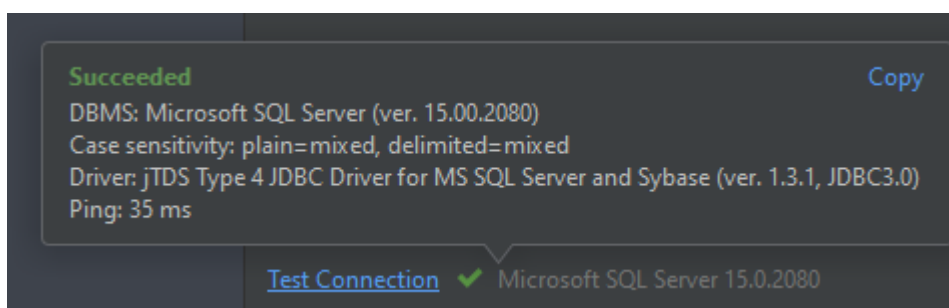
Host: postgresql.mandela.ac.za

User: WRAP301User

Password: 1 (I know Dieter used 1234, but that doesn't seem to work now)

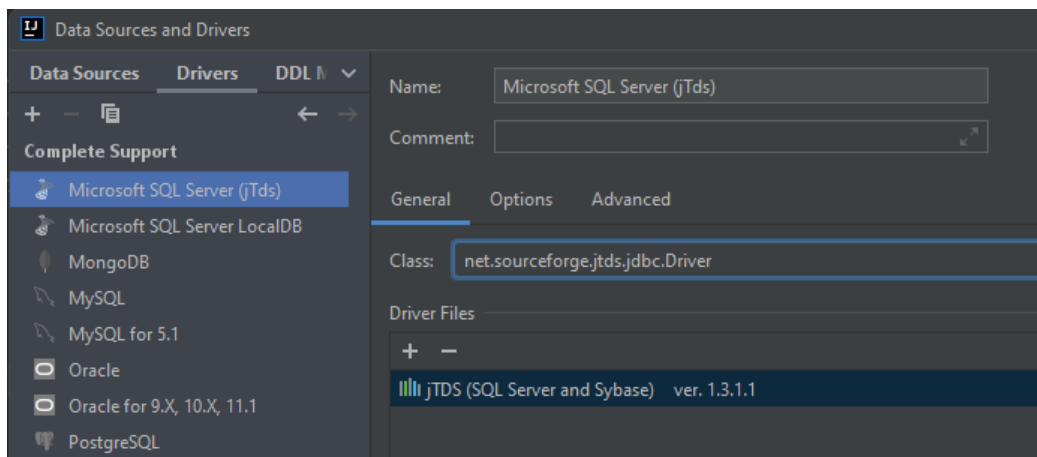
URL: jdbc:jtds:sqlserver://postgresql.mandela.ac.za/WRAP301;instance=WRR

When you press the Test Connection button, it should succeed (if it doesn't, then make sure you are connected to the internet, that you are connected to the VPN and maybe also make sure there's no loadshedding on campus?)

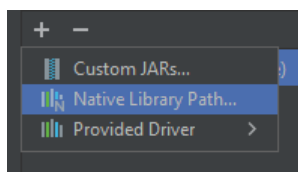


Once your connection succeeded, it means you must now update the code to make use of this new driver.

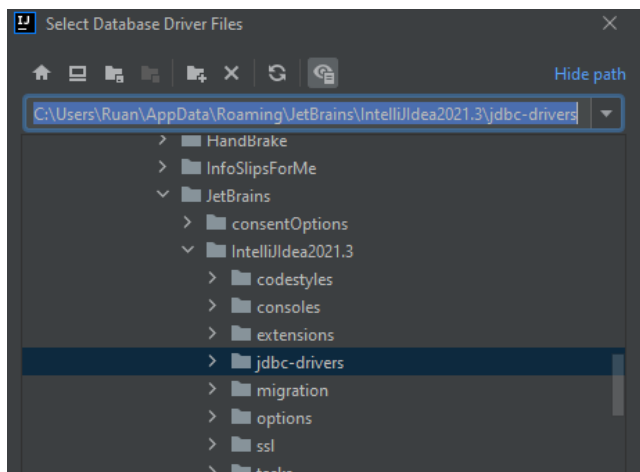
The first thing we want to do is add the jTds driver to the project's library. So in the same window, we swap to the Drivers tab and choose the applicable one (take note of the class... we will use it later)



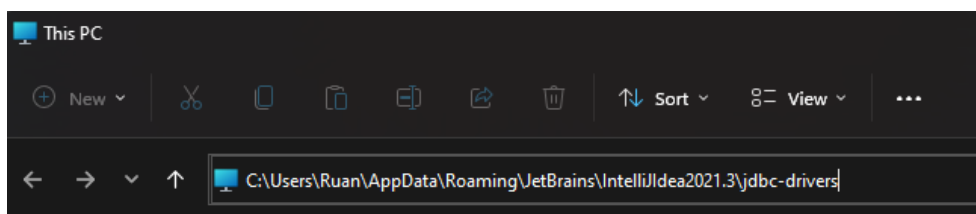
Press the + button and choose Native Library Path



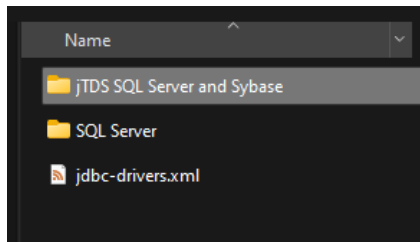
It will give you the path that the driver downloaded to. So copy this path (as we want to make a copy of the driver for this project's files)



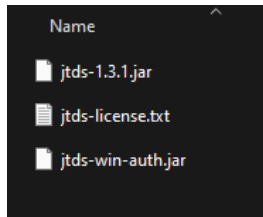
Then paste the path into File Explorer.



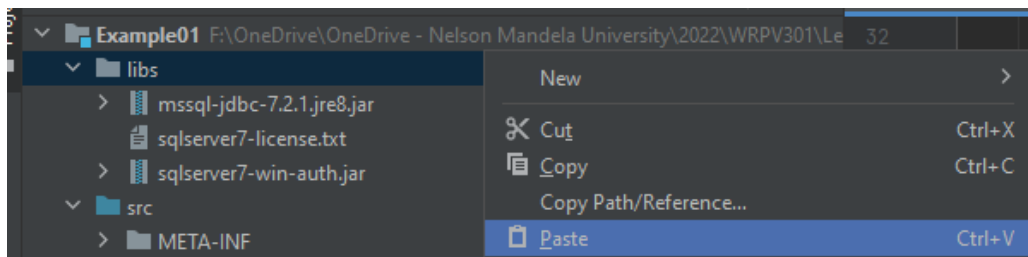
It should look something like:



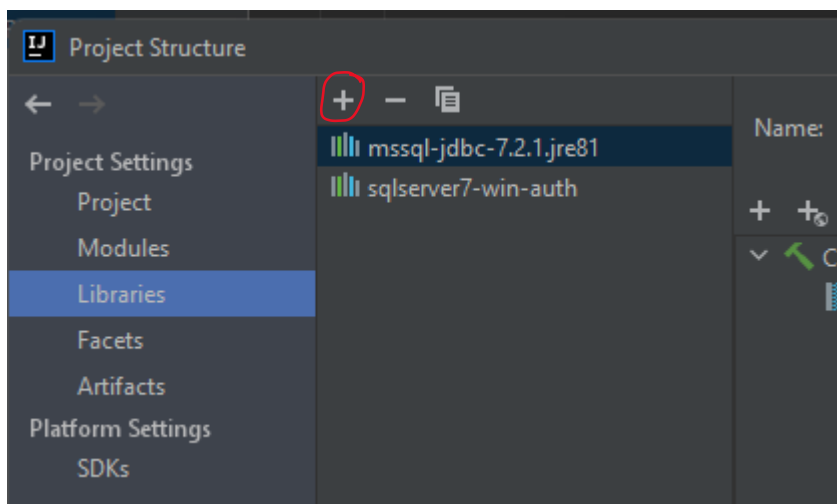
Open the top folder (and the inner folder with the version number) and copy the files inside.



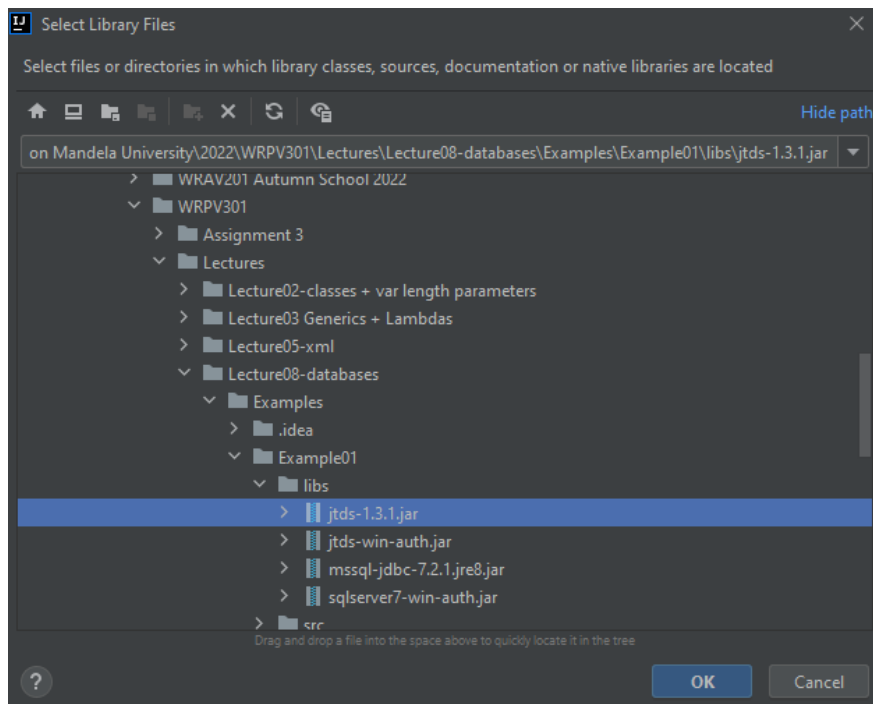
Now paste it inside the libs folder (remember: you will need to make this folder yourself for the prac task)



Once the driver files are in the libs folder, we will add it as a library to our project. So we open the module settings (view > Open Module settings), swap to the Libraries tab and click on the +



Choose the correct driver (jtds).



Finally, we must change some lines of code inside the connectToDB method. The line where we check that the driver is present, must change to use the jTds one. The connection string will also change and then the password must also be 1.

```
42 public void connectToDB() {
43     System.out.println("Establishing connection to database...");
44
45     System.out.println("    Loading JDBC driver for MS SQL Server database...");
46     try {
47         Class.forName("com.microsoft.sqlserver.jdbc.SQLServerDriver");
48     } catch (Exception e) {
49         System.out.printf("    Unable to load JDBC driver... '%s'\n", e.getMessage());
50         return;
51     }
52
53     System.out.println("    Use driver to connect to MS SQL Server (OPENBOX\\WRR)...");
54     if (true) {
55         try {
56             System.out.println("        Locate database to open (using connection string)...");
57
58             String connectionString = "jdbc:sqlserver://OPENBOX\\WRR;databaseName=WRAP301";
59             System.out.println("        Connection string = " + connectionString);
60
61             // create connection to DB, including username & password
62             // NEVER, EVER, include a username and password in your code!!!!
63             con = DriverManager.getConnection(connectionString, user: "WRAP301User", password: "1234");
64
65             // create statement object for manipulating DB
66             stmt = con.createStatement(ResultSet.TYPE_SCROLL_SENSITIVE, ResultSet.CONCUR_UPDATABLE);
67         } catch (Exception e) {
68             System.out.printf("        Unable to connect to DB... '%s'\n", e.getMessage());
69         }
70     } else {
```

Below are the new lines of code (in case you want to copy paste it 😊)

```
Class.forName("net.sourceforge.jtds.jdbc.Driver");
```

```
String connectionString = "jdbc:jtds:sqlserver://postgresql.mandela.ac.za/WRAP301;instance=WRR";
```

```
con = DriverManager.getConnection(connectionString, "WRAP301User", "1");
```

The code must then look like:

```
42 public void connectToDB() {
43     System.out.println("Establishing connection to database...");
44
45     System.out.println("    Loading JDBC driver for MS SQL Server database...");
46     try {
47         Class.forName("net.sourceforge.jtds.jdbc.Driver");
48     } catch (Exception e) {
49         System.out.printf("    Unable to load JDBC driver... '%s'\n", e.getMessage());
50         return;
51     }
52
53     System.out.println("    Use driver to connect to MS SQL Server (OPENBOX\\WRR)...");
54     if (true) {
55         try {
56             System.out.println("    Locate database to open (using connection string)...");
57
58             String connectionString = "jdbc:jtds:sqlserver://postgresql.mandela.ac.za/WRAP301;instance=WRR";
59             System.out.println("    Connection string = " + connectionString);
60
61             // create connection to DB, ubcluding username & password
62             // NEVER, EVER, include a username and password in your code!!!!
63             con = DriverManager.getConnection(connectionString, user: "WRAP301User", password: "1");
64
65             // create statement object for manipulating DB
66             stmt = con.createStatement(ResultSet.TYPE_SCROLL_SENSITIVE, ResultSet.CONCUR_UPDATABLE);
67         } catch (Exception e) {
68             System.out.printf("    Unable to connect to DB... '%s'\n", e.getMessage());
69         }
70     } else {
```

Run the program and it should work 😊.